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E-461 SUBSTATION MONITORING SYSTEM

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to site, installation, commissioning and testing of substation monitoring system in accordance with Part 4, Subsection 1.2 Summary of Work.

Monitoring system is available for the existing substation. The Contractor shall provide or extent (whichever applicable) for the monitoring system for the extended substation and existing diameter 5. Extended substation and existing diameter 5 substation shall be able to be monitored from the new power plant control room, substation control room and the load dispatch centre.

The extended substation and existing diameter 5 monitoring system shall include annunciator system, sequential event recording system, data acquisition system, and fault recording system.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Section 2.5 Codes and Standards.

3. OPERATING AND DESIGN CONDITIONS

The Substation is located in a tropical climate along the seacoast. Refer to Part 4, Subsection 1.1, 2.14 and 2.16 for meteorological, seismic and environmental data of the site.

4. DETAILED REQUIREMENTS

4.1 General Requirements

Whenever a common input is required by the annunciator and the event recorder, separate field contact shall be provided for each function unless the Contractor can prove to the Owner's satisfaction that the sharing of the contact will not compromise the overall reliability of the system. Each input shall be optically buffered with buffers capable of 220 V DC on either side. A maximum of 50 micro second delays is allowed from initiating contact closure to sensing by all monitoring equipment.

Normally-open or normally-closed signal inputs on the equipment may be moni-

tored which will close or open respectively on alarm conditions. Each field contact shall be independently wired to the inputs cubicles. Any required parallel or series connection of field contacts will be made at the input cubicles, preferably in logic form. All points shall be field programmable to operate on either a normal-open or closed field contact.

Field contacts shall be dry contacts.

Solid State Plug in Cards.

- (1) Silicon solid-state component shall be used throughout the system equipment.
- (2) All printed circuit cards shall be epoxy-fiberglass boards and all connectors shall be gold-plated.
- (3) Solid-state circuits shall be of the printed board plug-in type. All components shall be soldered in place to eliminate the chance of malfunction due to poor connection.
- (4) Plug-in design and conduction shall be followed concerning light boxes, as well as modular circuit cards. Each input contact shall be independently wired to the terminal blocks.
- (5) Circuits shall be arranged so that individual point card can be safely replaced without interrupting of the rest of the system.

Ground Isolation and Indication

- (1) Ground isolation switches and ground indicators shall be provided.
- (2) A ground detector to detect the presence of a ground on the input leads shall be provided. A common annunciator point shall be provided to annunciate detection of a ground. The ground detector will lock until manually reset.
- (3) Circuitry shall be arranged to detect grounds on one (1) side of normally-open field contacts. Ground detection shall include power supply(s) and system wiring monitoring.

Test Facilities

- (1) Equipment furnished under this Technical Specification shall be designed and arranged to facilitate periodic testing, calibration and maintenance.

- (2) The Contractor shall furnish all test equipment or tools required for normal maintenance, testing and calibration.

4.2 Annunciator System

4.2.1 General

Beside scanning and alarming as part of the monitoring system specified in Technical Specification I-200, the substation annunciator system shall also be provided. Supply has to be made according to the existing system.

- (1) The annunciator is a modular solid-state type. The system shall be completed with visual indicators consisting of back lighted engraved windows.
- (2) Circuits shall be arranged so that individual point cards or lamps can be safely replaced without interrupting the rest of the system.
- (3) The annunciator shall be furnished complete with all points wired and auxiliary devices, signal horn, “Acknowledge”, “Reset”, and “Test” push buttons installed in the control panels.
- (4) Each alarm point shall be capable of sounding the audible signal independent of the status of all other alarm points in the system. Each alarm point shall be controlled by common push button (Acknowledge, Test, Reset), independent of the status of all other alarm points in the system.
- (5) The test pushbutton shall be provided for a complete system test by simulating an input to the input logic cards.

4.2.2 Power Supply

The power source, provided as a separate system, of 220 volts ungrounded DC. The Contractor shall supply a static power supply, for ten (10) hours loss of AC as described in Part 4, Technical Specification E-450. The annunciator positive will be used for the trouble contact (field contacts). The Contractor shall provide an alarm relay which will monitor the annunciator power supply to give output for remote indication of power supply trouble. Circuit breakers shall be provided for power supplies.

4.2.3 The Sequence of Operation

The sequence of operation of the annunciator points shall be as follows:

STATE	FIELD CONTACTS	ANUNCIATE LIGHTS	HORN
1. Normal	Normal	Off	Off
2. Alarm	Abnormal	Fast Flash	On
3. Acknowledge	Abnormal	On	Off
4. Return to Normal	Normal	Slow Flash	Off
5. Reset	Normal	Off	Off
6. Test	Normal	On	On

4.2.4 Annunciator Windows

Windows shall be translucent white background with black engraved letters.

The windows shall be of a snap-in contraction to permit the lamp replacement from the front.

Each window shall be provided with two (2) lamps. Loss of one (1) of the lamps shall not interfere with functioning of the other lamp. Brightness of the windows shall be sufficient to provide ample contrast with illumination of 100 to 200 Lux.

The alarm list, window engraving list, engraving size and window arrangement shall be subject to the Owner's review.

4.2.5 Window Arrangements

Annunciator window arrangement is functionally independent per bay with individual push buttons on each control panel. Window quantities shall be sufficient to cover all alarm in that bay and sufficient spares shall be provided.

4.3 Sequential Event Recording (SER)

4.3.1 General

Unless otherwise specified herein, the Contractor's supply shall be conform with the existing SER equipment for integration of the new scope.

Sequential event recording shall be provided to monitor the substation events and shall provide a printed, English language record of field alarms, switching equipment status and all the important events with their time of occurrence, for later analysis.

Field alarm contact voltage shall be 220 V DC (nominal) as supplied by the Con-

tractor's separate field contact power sources for inputs other than the approved annunciator buffered inputs.

4.3.2 Function

The sequential event recording shall function to record an event as follows:

- (1) At the start of each sequence of event printouts, the date shall be printed.
- (2) True time shall be printed with each event, i.e. actual time accurately to the nearest second and millisecond.
- (3) The events shall be logged into memory in sequence of operation if the operations are spaced at least 0.001 second apart, up to capacity of the memory. The events occurring in intervals less than 0.001 second shall be accurately recorded but not necessarily in sequence.
- (4) The printout shall be in English language (alphanumeric). The English language description for each point shall be field programmable from the plug-in keyboard.
- (5) An alarm summary function shall be provided. A button located on the printer, when pressed, shall initiate a printout of all points currently in alarm condition. Should events occur during this operation, the events shall be stored and printed as a standard sequence of events printout.
- (6) A printout shall be provided to identify the beginning of each 24-hour day and date.

If an event is printed, the alarm condition and the return to normal condition must be printed, except certain points do not require printing of the return to normal condition. On the points for which return to normal printing is inhibited, the return to normal condition shall not be placed into sequential memory. Event recorder shall be programmable to inhibit return to normal printing of any print.

Proper functioning of each power supply shall be monitored, and failure due to any cause, shall close a contact which will give alarm.

The following circuitry and functions shall be provided:

- (1) The logic equipment shall automatically check the printer each hour.
- (2) The logic equipment shall be provided with plug-in type sequential memory facilities. Each such memory facility shall be released for further use upon

printing of appoint held in memory. Alarms being held in memory shall advance to void all vacancies and allow additional alarm to enter memory.

- (3) Full operational test on all inputs shall be provided. When the printer test push button is operated, only those points which are faulty shall print out. The equipment shall be required to distinguish between field contact in the alarm position and faulty inputs inside the recorder.
- (4) Alarms shall be provided for all recorder system malfunctions. A lamp panel detailing the recorder malfunction shall be provided inside the recorder cabinet.
- (5) The alarm lamps shall be hand reset by a master alarm reset push button located on the lamp panel.
- (6) The event recorder shall be provided with a digital clock for printing the time of the events. This clock shall be synchronized to the master clock specified in Technical Specification I-200.

The system shall be provided with a minimum monitoring capacity of 800 events. The Contractor shall, with his proposal, include a list of events he proposes to monitor. This list shall be subject to the Owner's review until the completion of the drawing review. The following list provides a general guideline for the type of events to be monitored:

- (1) All annunciator alarm including reset.
- (2) All 500 kV Circuit breaker activities.
- (3) All 500 kV protective relay actions including reset, for relaying, plant interfaces, etc.
- (4) All remote protective actions and resets.
- (5) All 500 kV disconnecting and grounding switches.

4.3.3 Power Supplies

The normal power feed shall be from the 220 volts DC battery. Each Power supply shall be capable of continuously supplying the maximum power demand.

A 220 volt AC supply shall be connected as the back-up source for the DC supply.

4.3.4 Interconnections and Programming

Plugs and cables shall be provided to connect programming keyboard for initiali-

zation and revising alarm input function and English language description.

4.3.5 Printers

Paper advance shall occur with the printouts and not be continuous drive. Manual advance shall be provided. Memory shall be permanent until recorder.

The printer shall provide a record of all operations, showing the time with 0.001-second minimum discrimination, type of operated (normal to abnormal or reverse), and identification of the point. Point identifications shall consist of a unique number and an English language (alphanumeric) description.

The printer motor shall not run continuously but shall start when the sequential event recorder logic requires a printout.

4.4 Fault Recording

4.4.1 Function

The fault recorder system shall continuously monitor system voltage and current waveform and shall produce a paper record of the waveforms when initiated from one (1) of the fault sensors provided with the fault recorder. This paper record will be used for a post fault analysis along with the sequential event recorder printout shall also be capable of being initiated through an external contact closure.

Each fault recorder shall include 256 milliseconds of prefault waveform, waveforms during the fault condition and post fault waveforms up to the setting of the operation limiter.

Sufficient voltage and current waveforms shall be monitored to provide a complete post fault analysis. As a minimum the following waveform shall be monitored:

- (1) Individual phase and neutral current of generators, and transformer feeds.
- (2) Individual phase and neutral voltage waveform of transformer circuit at 500 kV, etc.

Sufficient starting element shall be provided to detect each fault or system disturbance. As minimum, the following fault sensors shall be provided:

- (1) Zero sequence current of generator
- (2) Negative sequence 50 Hz current of generator.
- (3) Under voltage at substation bus bars.

The station code, date and time of printout shall be written on the recording paper by a non contacting device.

The fault recording system shall be equipped with an operation limiter to stop the record after an adjustable period of one (1) to ten (10) seconds. If another disturbance is detected, the overrun timer shall reset and fault recorder shall record the second disturbance and the corresponding post fault data.

4.4.2 General

The fault recorder shall be the direct type utilizing a light beam and paper or equal. The paper capacity shall be at least 65 meters. A paper take-up reel with capacity of at least 65 meters shall be furnished. The fault recorder shall be sealed to prevent dust from getting into moving parts.

All channels whether designated as “spare” or intended for use at this time, shall be completely wired out for connection.

The fault recorder starting time shall be within two (2) ms from the time of occurrence of a fault situation. After attaining proper speed, this speed shall be maintained. The fault recorder shall be capable of dual speed operation.

The galvanometers light source and optical system shall be capable of producing a full contrast trace of the recording paper within two (2) ms after the occurrence of a fault situation.

Initiation of the trace shall be from internal initiating devices or external dry contacts. The fault recorder shall shut down if the paper supply runs out.

The fault recorder shall have the following alarm indications on the system panel:

- (1) Low paper alarm (less than three reclosing sequence operation times).
- (2) Out of paper alarm.
- (3) Fault recording system failure.

- (4) Fault recorder operation.
- (5) DC power supply failure.
- (6) Lamp failure.
- (7) Clock and control circuitry failure.

The fault recorder shall be provided with a digital clock, which shall be synchronized with the master clock specified in Technical Specification I-200.

Each printout shall be identified with date, time and a three-digit code for station identification.

4.4.3 Input Signals

Input conditioners shall be furnished for all analog recording channels. All channels shall be capable of accepting either AC current (0-5 amperes nominal) or AC voltage (0-220 volts nominal). The overload capability shall be minimum 200 amperes for one second and 1,000 volts for one second, respectively. All galvanometers shall have easily accessible level controls.

The input conditioners shall also furnish isolation in the oscillograph recording system of 2,500 volts rms channel to channel and 2,500 volts rms channel to ground.

Frequency response of all input conditioning equipment shall be DC to 1,500 Hz minimum.

All shunts used within the oscillograph recording system shall have a continuous current rating of at least ten (10) amperes rms and a one (1) second rating of at least 200 amperes rms. All inputs shall be interchangeable between current and potential without having to install any additional devices.

Knife switches and all necessary fusing for the power supply equipment power shall be furnished, installed and wired.

All analog inputs to the fault recorder system shall be provided with test switches for isolation of the recording system from the current and potential transformers.

5. PERFORMANCE TESTING AND GUARANTEE

Type tests report on identical equipment shall be submitted.

Routine tests in the manufacturer's factory shall include, but not be limited to,

- (1) Complete system functional test including simulation of inputs and outputs covering the system.
- (2) Heat run tests of all electronic and solid-state hardware.
- (3) Tests designed to identify faults in programmed systems.
- (4) Calibration checks on samples of each production batch of instruments which do not have project specified settings.
- (5) Calibration and adjustment to all instruments with project specific settings.

6. QUALITY ASSURANCE/QUALITY CONTROL

The quality assurance/quality control of substation monitoring system shall conform to the requirements included in Part 4 Section 2. Subsection 2.7. The procedures shall include, but not be limited to, the followings:

- (1) Calibration and accuracy verifications tests.
- (2) Type tests for all test performed for this Contract.
- (3) Functional tests.
- (4) Fault identifications tests.
- (5) Surge withstand tests and control wiring tests.

The quality assurance/quality control documentation shall include, but not be limited to, the followings:

- (1) Shop Test Report.
- (2) Test report for all tests specified in Clause 5 above.
- (3) All other documentation required by applicable codes and standards.

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